

$$\begin{aligned}
& \frac{1}{16\pi^2} \left(\frac{1}{144} g_1 c_Y \frac{1}{m_s^2} (3 s_Y^2 \overline{y_d^{pr}} (y_d^{pi2} y_d^{i1r} + 6 y_d^{pr} y_d^{i1i2}) + y_d^{pi2} \overline{y_u^{pr}} y_u^{i1r} (11 c_Y^2 + 3 s_Y^2)) - \right. \\
& \frac{5}{24} g_1 c_Y s_Y^2 y_d^{pi2} \overline{y_u^{pr}} y_u^{i1r} LF_{1,2}[m_\Phi] - \frac{1}{54} c_Y g_1^3 y_d^{i1i2} LF_{2,1,0}[m_1, m_d^{-i2}] + \\
& \frac{1}{27} c_Y g_1^3 y_d^{i1i2} LF_{3,1,-1}[m_1, m_d^{-i2}] - \frac{1}{54} c_Y g_1^3 y_d^{i1i2} LF_{4,1,-2}[m_1, m_d^{-i2}] + \\
& \frac{1}{432} c_Y g_1^3 y_d^{i1i2} LF_{2,1,0}[m_1, m_q^{i1}] - \frac{1}{216} c_Y g_1^3 y_d^{i1i2} LF_{3,1,-1}[m_1, m_q^{i1}] + \\
& \frac{1}{432} c_Y g_1^3 y_d^{i1i2} LF_{4,1,-2}[m_1, m_q^{i1}] + \frac{1}{16} g_1 c_Y g_2^2 y_d^{i1i2} LF_{2,1,0}[m_2, m_q^{i1}] - \\
& \frac{1}{8} g_1 c_Y g_2^2 y_d^{i1i2} LF_{3,1,-1}[m_2, m_q^{i1}] + \frac{1}{16} g_1 c_Y g_2^2 y_d^{i1i2} LF_{4,1,-2}[m_2, m_q^{i1}] - \\
& \frac{2}{9} g_1 c_Y g_3^2 y_d^{i1i2} LF_{2,1,0}[m_3, m_d^{-i2}] + \frac{4}{9} g_1 c_Y g_3^2 y_d^{i1i2} LF_{3,1,-1}[m_3, m_d^{-i2}] - \\
& \frac{2}{9} g_1 c_Y g_3^2 y_d^{i1i2} LF_{4,1,-2}[m_3, m_d^{-i2}] + \frac{1}{9} g_1 c_Y g_3^2 y_d^{i1i2} LF_{2,1,0}[m_3, m_q^{i1}] - \\
& \frac{2}{9} g_1 c_Y g_3^2 y_d^{i1i2} LF_{3,1,-1}[m_3, m_q^{i1}] + \frac{1}{9} g_1 c_Y g_3^2 y_d^{i1i2} LF_{4,1,-2}[m_3, m_q^{i1}] - \\
& \frac{1}{12} g_1 c_Y \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} LF_{2,1,0}[\tilde{\mu}, m_d^{-r}] + \frac{1}{24} g_1 c_Y \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} LF_{3,1,-1}[\tilde{\mu}, m_d^{-r}] + \\
& \frac{1}{24} g_1 c_Y \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} LF_{4,1,-2}[\tilde{\mu}, m_d^{-r}] + \\
& \frac{1}{12} g_1 c_Y \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} LF_{2,1,0}[\tilde{\mu}, m_q^p] + \frac{1}{12} g_1 c_Y \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} LF_{3,1,-1}[\tilde{\mu}, m_q^p] - \\
& \frac{1}{6} g_1 c_Y \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} LF_{4,1,-2}[\tilde{\mu}, m_q^p] + \frac{1}{6} g_1 c_Y y_d^{pi2} \overline{y_u^{pr}} y_u^{i1r} LF_{2,1,0}[\tilde{\mu}, m_u^r] - \\
& \frac{5}{24} g_1 c_Y y_d^{pi2} \overline{y_u^{pr}} y_u^{i1r} LF_{3,1,-1}[\tilde{\mu}, m_u^r] + \frac{1}{24} g_1 c_Y y_d^{pi2} \overline{y_u^{pr}} y_u^{i1r} LF_{4,1,-2}[\tilde{\mu}, m_u^r] + \\
& \frac{1}{216} m_1 g_1^3 (-c_Y a_d^{i1i2} + s_Y \tilde{\mu} y_d^{i1i2}) LF_{2,1,1,0}[m_1, m_d^{-i2}, m_q^{i1}] + \\
& \frac{1}{144} m_1 g_1^3 (-c_Y a_d^{i1i2} + s_Y \tilde{\mu} y_d^{i1i2}) LF_{2,2,1,-1}[m_1, m_d^{-i2}, m_q^{i1}] + \frac{1}{216} m_1 g_1^3 \\
& (c_Y a_d^{i1i2} - s_Y \tilde{\mu} y_d^{i1i2}) LF_{3,1,1,-1}[m_1, m_d^{-i2}, m_q^{i1}] - \frac{1}{18} c_Y g_1^3 y_d^{i1i2} LF_{1,1,1,0}[m_1, m_d^{-i2}, \tilde{\mu}] + \\
& \frac{1}{9} c_Y g_1^3 y_d^{i1i2} LF_{2,1,1,-1}[m_1, m_d^{-i2}, \tilde{\mu}] + \frac{1}{18} m_1 s_Y \tilde{\mu} g_1^3 y_d^{i1i2} LF_{2,1,1,0}[m_1, m_d^{-i2}, \tilde{\mu}] - \\
& \frac{1}{18} c_Y g_1^3 y_d^{i1i2} LF_{3,1,1,-2}[m_1, m_d^{-i2}, \tilde{\mu}] - \frac{1}{18} m_1 s_Y \tilde{\mu} g_1^3 y_d^{i1i2} LF_{3,1,1,-1}[m_1, m_d^{-i2}, \tilde{\mu}] + \\
& \frac{1}{144} m_1 g_1^3 (c_Y a_d^{i1i2} - s_Y \tilde{\mu} y_d^{i1i2}) LF_{2,2,1,-1}[m_1, m_q^{i1}, m_d^{-i2}] + \\
& \frac{1}{72} c_Y g_1^3 y_d^{i1i2} LF_{1,1,1,0}[m_1, m_q^{i1}, \tilde{\mu}] - \frac{1}{36} c_Y g_1^3 y_d^{i1i2} LF_{2,1,1,-1}[m_1, m_q^{i1}, \tilde{\mu}] - \\
& \frac{1}{72} m_1 s_Y \tilde{\mu} g_1^3 y_d^{i1i2} LF_{2,1,1,0}[m_1, m_q^{i1}, \tilde{\mu}] + \frac{1}{72} c_Y g_1^3 y_d^{i1i2} LF_{3,1,1,-2}[m_1, m_q^{i1}, \tilde{\mu}] + \\
& \frac{1}{72} m_1 s_Y \tilde{\mu} g_1^3 y_d^{i1i2} LF_{3,1,1,-1}[m_1, m_q^{i1}, \tilde{\mu}] - \frac{1}{72} c_Y g_1^3 y_d^{i1i2} LF_{2,2,1,-2}[m_1, \tilde{\mu}, m_d^{-i2}] - \\
& \frac{1}{72} m_1 s_Y \tilde{\mu} g_1^3 y_d^{i1i2} LF_{2,2,1,-1}[m_1, \tilde{\mu}, m_d^{-i2}] - \frac{1}{144} c_Y g_1^3 y_d^{i1i2} LF_{2,2,1,-2}[m_1, \tilde{\mu}, m_q^{i1}] - \\
& \frac{1}{144} m_1 s_Y \tilde{\mu} g_1^3 y_d^{i1i2} LF_{2,2,1,-1}[m_1, \tilde{\mu}, m_q^{i1}] + \frac{1}{8} g_1 c_Y g_2^2 y_d^{i1i2} LF_{1,1,1,0}[m_2, m_q^{i1}, \tilde{\mu}] - \\
& \frac{1}{4} g_1 c_Y g_2^2 y_d^{i1i2} LF_{2,1,1,-1}[m_2, m_q^{i1}, \tilde{\mu}] - \frac{1}{8} g_1 m_2 s_Y \tilde{\mu} g_2^2 y_d^{i1i2} LF_{2,1,1,0}[m_2, m_q^{i1}, \tilde{\mu}] + \\
& \frac{1}{8} g_1 c_Y g_2^2 y_d^{i1i2} LF_{3,1,1,-2}[m_2, m_q^{i1}, \tilde{\mu}] + \frac{1}{8} g_1 m_2 s_Y \tilde{\mu} g_2^2 y_d^{i1i2} LF_{3,1,1,-1}[m_2, m_q^{i1}, \tilde{\mu}] - \\
& \frac{1}{16} g_1 c_Y g_2^2 y_d^{i1i2} LF_{2,2,1,-2}[m_2, \tilde{\mu}, m_q^{i1}] - \frac{1}{16} g_1 m_2 s_Y \tilde{\mu} g_2^2 y_d^{i1i2} LF_{2,2,1,-1}[m_2, \tilde{\mu}, m_q^{i1}] + \\
& \frac{1}{9} g_1 m_3 g_3^2 (c_Y a_d^{i1i2} - s_Y \tilde{\mu} y_d^{i1i2}) LF_{2,1,1,0}[m_3, m_d^{-i2}, m_q^{i1}] + \\
& \frac{1}{6} g_1 m_3 g_3^2 (c_Y a_d^{i1i2} - s_Y \tilde{\mu} y_d^{i1i2}) LF_{2,2,1,-1}[m_3, m_d^{-i2}, m_q^{i1}] + \\
& \frac{1}{9} g_1 m_3 g_3^2 (-c_Y a_d^{i1i2} + s_Y \tilde{\mu} y_d^{i1i2}) LF_{3,1,1,-1}[m_3, m_d^{-i2}, m_q^{i1}] + \\
& \frac{1}{6} g_1 m_3 g_3^2 (-c_Y a_d^{i1i2} + s_Y \tilde{\mu} y_d^{i1i2}) LF_{2,2,1,-1}[m_3, m_q^{i1}, m_d^{-i2}] + \\
& \frac{1}{16} g_1 \tilde{\mu} y_d^{pi2} y_u^{i1r} (-s_Y \overline{a_u^{pr}} + \tilde{\mu} c_Y \overline{y_u^{pr}}) LF_{2,2,1,-1}[m_q^p, \tilde{\mu}, m_u^r] + \\
& \frac{1}{16} g_1 \tilde{\mu} y_d^{pi2} y_u^{i1r} (s_Y \overline{a_u^{pr}} - \tilde{\mu} c_Y \overline{y_u^{pr}}) LF_{2,2,1,-1}[m_u^r, \tilde{\mu}, m_q^p] + \\
& \frac{1}{36} c_Y g_1^3 y_d^{i1i2} LF_{2,1,1,-1}[\tilde{\mu}, m_1, m_d^{-i2}] + \frac{1}{18} m_1 s_Y \tilde{\mu} g_1^3 y_d^{i1i2} LF_{2,1,1,0}[\tilde{\mu}, m_1, m_d^{-i2}] + \\
& \frac{1}{36} c_Y g_1^3 y_d^{i1i2} LF_{3,1,1,-2}[\tilde{\mu}, m_1, m_d^{-i2}] + \frac{1}{36} m_1 s_Y \tilde{\mu} g_1^3 y_d^{i1i2} LF_{3,1,1,-1}[\tilde{\mu}, m_1, m_d^{-i2}] + \\
& \frac{1}{72} c_Y g_1^3 y_d^{i1i2} LF_{2,1,1,-1}[\tilde{\mu}, m_1, m_q^{i1}] - \frac{1}{72} m_1 s_Y \tilde{\mu} g_1^3 y_d^{i1i2} LF_{2,1,1,0}[\tilde{\mu}, m_1, m_q^{i1}] - \\
& \frac{1}{36} c_Y g_1^3 y_d^{i1i2} LF_{3,1,1,-2}[\tilde{\mu}, m_1, m_q^{i1}] - \frac{1}{36} m_1 s_Y \tilde{\mu} g_1^3 y_d^{i1i2} LF_{3,1,1,-1}[\tilde{\mu}, m_1, m_q^{i1}] + \\
& \frac{1}{8} g_1 c_Y g_2^2 y_d^{i1i2} LF_{2,1,1,-1}[\tilde{\mu}, m_2, m_q^{i1}] - \frac{1}{8} g_1 m_2 s_Y \tilde{\mu} g_2^2 y_d^{i1i2} LF_{2,1,1,0}[\tilde{\mu}, m_2, m_q^{i1}] - \\
& \frac{1}{4} g_1 c_Y g_2^2 y_d^{i1i2} LF_{3,1,1,-2}[\tilde{\mu}, m_2, m_q^{i1}] - \frac{1}{4} g_1 m_2 s_Y \tilde{\mu} g_2^2 y_d^{i1i2} LF_{3,1,1,-1}[\tilde{\mu}, m_2, m_q^{i1}] + \\
& \frac{5}{24} g_1 \tilde{\mu} y_d^{pi2} y_u^{i1r} (s_Y \overline{a_u^{pr}} - \tilde{\mu} c_Y \overline{y_u^{pr}}) LF_{2,1,1,0}[\tilde{\mu}, m_q^p, m_u^r] + \\
& \left. \frac{1}{24} g_1 \tilde{\mu} y_d^{pi2} y_u^{i1r} (s_Y \overline{a_u^{pr}} - \tilde{\mu} c_Y \overline{y_u^{pr}}) LF_{3,1,1,-1}[\tilde{\mu}, m_q^p, m_u^r] \right)
\end{aligned}$$